LIBBY — PATIENT/CAREGIVER TRACK

pancreatic-mets-kras-g12d-basal-arid1a-k9r3

Plain-language summary for patient and caregiver

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This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

Libby read your tumor's genetic profile and gathered the treatment options that aim at the specific features driving this cancer. It ranked them and wrote down the promise and the problems for each one. This is the plain-language version of that work, written for you and the people helping you decide rather than for a specialist.

Before you go further, one thing to hold onto. This page only covers treatments that target the molecular features found in your tumor. The standard chemotherapy that almost everyone with this diagnosis is offered first is a separate conversation with your oncologist. Some of the trials below build on that chemotherapy, so it comes up as a backbone, but it is not ranked here and your care team owns that decision.

What we know about your cancer

You have pancreatic cancer of the ductal adenocarcinoma type, and it has spread. There is a large mass in the pancreas itself and several tumors bigger than a centimeter in the liver. A gene-expression test placed your tumor in the "basal-like" group, which tends to behave more aggressively than the other main subtype, and that plays into which chemotherapy partner your team would choose.

The genetic testing turned up a handful of things that shape the options:

- A change in a gene called KRAS, the G12D version. This is the main engine of the cancer and the most useful target found. Most of the options below aim straight at it.
- A loss-of-function change in a gene called ARID1A. This is a second, separate target, less established but real.
- A single change in a DNA-repair gene called MLH1. On its own, that can look like the kind of cancer that responds to immune-unleashing drugs. The rest of your profile points the other way, though: the tumor is "microsatellite stable" and its overall mutation count is on the lower side. So immune-checkpoint drugs by themselves are probably low-yield here. A simple tissue test (MMR protein staining, sometimes called IHC) settles the question, and it is worth running before anyone reaches for that kind of drug.
- A few other changes (in SMAD4, TP53, and some incidental genes) that mostly speak to prognosis rather than pointing at a drug. One of them, in a gene called ASXL1, may actually be coming from your blood cells rather than the tumor. A paired blood test can sort that out so it is not mistaken for a tumor target.

Your markers were read from a blood-based test, a liquid biopsy. Most of the trials will want the KRAS finding confirmed on an actual tissue sample before they enroll you, so getting a tissue biopsy looked at is a practical first errand, and the same tissue can run the MMR stain at the same time.

What you told us matters most

You did not give us specific goals or priorities, so we set the balance between fighting hard and going easy on side effects right in the middle. You did tell us you are open to clinical trials, and that matters a great deal here, because every targeted option for this cancer is currently available only through a trial.

One honest flag: you are 79. In someone's late seventies, the side-effect cost of a treatment usually weighs heavier than a neutral setting captures. We have not tilted the ranking on a preference you did not state, but it comes up repeatedly below, because the best-studied option and the gentlest option are not the same drug. If you and your oncologist decide that being easier on your body is the priority, the order near the top could reasonably shift. That is worth saying out loud in the room.

The options

These are ranked, but the gaps among the first three are narrow and partly a matter of what you value. Further down, the options get more uncertain, and the last few are on the page as things that were weighed rather than things to reach for now.

Option 1 — daraxonrasib (a KRAS-blocking pill)

This is the option to discuss first. It is a pill that blocks the KRAS protein driving your cancer, and it covers the broader RAS family, not just the single G12D spot. Of everything on this page, it has the strongest evidence behind it: a large randomized trial of about 500 people, where some got the drug and some got chemotherapy, so the comparison is a real one.

In that trial, run in people who had already had one prior treatment, the results were striking. People on daraxonrasib lived a median of about 13 months, against about 7 months on chemotherapy. Put another way, at the point the researchers analyzed the data, the drug roughly cut the risk of dying by about 60% compared with chemo. The cancer also shrank in about a third of people, versus about one in nine on chemo, and the serious side effects were actually fewer than chemo caused in the same trial.

Here is the catch, and it matters. That trial was in people who had already had a first treatment. You would be starting this as your first treatment, and the trial testing it in that earlier setting (called RASolute 303) has not reported yet. So the benefit at your stage is expected, based on how the drug worked later on, but not yet proven. The survival numbers above also come from a mix of KRAS subtypes; your exact G12D version sat inside that group but was not analyzed on its own.

The main side effects are a skin rash and mouth sores, severe in roughly one in seven people, plus the general side effects any cancer drug brings. For an older person, the rash and mouth-sore problem is the thing to watch, because managing them sometimes means cutting the dose, and a lower dose can mean less benefit. The trial team has a playbook for handling these. It is a pill you take continuously, it fits your openness to trials, and the first-line trial accepts your KRAS type.

Option 2 — GFH375 (also written VS-7375), a KRAS pill given with an antibody called cetuximab

This is a second KRAS-blocking pill, and it is built to hit your exact G12D version. Of all the options here, it posted the loudest tumor-shrinkage number: in an early study of about 59 people with your exact mutation, the cancer shrank in roughly 41%, and it was held in check in nearly everyone. That is the highest single-drug response on your specific mutation anywhere on this page.

Two things keep it from the top spot. First, that study had no comparison group, it was run at a single center in China, and most of the people in it had already been through two or more prior treatments, while you would be starting fresh. A response rate from that kind of study is a weaker footing than Option 1's randomized survival result. Whether the 41% carries over to a first-line patient is genuinely unknown until the first-line group reports. Second, the trial you would enter pairs this pill with cetuximab, an antibody given by IV. The pill's own side effects (some low blood counts, diarrhea) have known management routes, but the two together have never had their own safety report in pancreatic cancer, so how the combination sits with a 79-year-old is uncharacterized. That combination piece is the open question.

It matches you on the mutation and on your openness to trials, and the first-line slot is recruiting now.

Option 3 — zoldonrasib (a more precisely targeted KRAS pill)

This is another KRAS-blocking pill built to hit your exact G12D mutation, and it is the gentlest drug on this whole page. In an early study of about 40 people with your mutation, the cancer shrank in about 30% and was held in check in about 80%. There were no severe (grade 4 or 5) side effects at the dose used; what did show up (nausea, some diarrhea, a little rash) was mostly mild.

So why third and not first? Because that 30% comes from an early study where everyone got the drug, with no comparison group, and only 40 people. We do not yet know how long the benefit lasts, because that has not been measured and reported. A response rate without any read on durability is weaker evidence than a randomized survival result. That is the open question here: real activity in your exact mutation, but staying power that nobody has measured yet.

For a 79-year-old, the tolerability is a real point in its favor. If going easy on your body is your priority, this is the drug a careful person would put forward first. The first-line trial (RASolute 305) adds it on top of chemotherapy, and the combined side effects of that pairing have not been published on their own, so the bone-marrow risk of the combination at your age is not yet a known quantity.

Option 4 — INCB161734 (another precise KRAS pill, given with chemo)

This option carries caveats. Here is the tradeoff to weigh. This is a fourth KRAS-blocking pill, from a different company, also aimed at your G12D mutation. On the response numbers it looks the strongest of the single agents: in an early study of about 41 people (all heavily pretreated), the cancer shrank in about 37%. On its own, its side effects stayed mild, and it notably did not cause the rash and mouth sores the others can.

The caveat is about how you would actually receive it. The first-line trial (DAWN-303) does not give this pill by itself; it stacks it on top of full chemotherapy. In the early combination data, about 40% of people had a serious drop in their infection-fighting white blood cells. That is the concern for a 79-year-old: the chemo-plus-pill combination hits the bone marrow harder, and there is no dedicated safety study of that combination yet. The favorable side-effect profile is real for the drug alone, but the version you would enroll in bundles in the chemotherapy and its risks.

On effectiveness, the 37% here and the 30% for Option 3 are close enough to be within the noise of two small studies, so this does not clearly beat Option 3. It sits a notch lower mainly because the route you would take carries that documented marrow hit.

Option 5 — HRS-4642 (a KRAS drug given by IV with chemotherapy)

This option carries caveats. Here is the tradeoff to weigh. This is a KRAS drug aimed at your G12D mutation, given by IV alongside chemotherapy, and its front-line number is the highest on this page. In an early study of 30 people starting treatment for the first time, the cancer shrank in about 63%. On paper that is the loudest first-line result here.

The tradeoff is severe, and there are two parts to it. The first is safety. In that study, about 90% of people had a serious (grade 3 or higher) side effect, mostly low blood counts, and there was one death from treatment in the wider safety group. For a 79-year-old, that is a heavy burden, and it runs against the general lean toward gentler treatment at your age. The second part is that the 63% number is hard to credit to the drug itself. It was given on top of chemotherapy, and chemotherapy alone shrinks these tumors in roughly a fifth of people. When this same drug was given by itself, without chemo, the response was only about 21%. So a large share of that 63% is the chemotherapy you would receive anyway, and there was no comparison group to separate the two. What the drug actually adds, on its own, is the open question.

There is also a practical wall right now: the trial that would offer this is not yet enrolling. It could become a real conversation for someone who put the highest possible response ahead of tolerability, but that is not where an unstated preference for a 79-year-old naturally points. It is on the page because Libby shows what was weighed, and this one was weighed and set aside for now on the strength of that safety profile.

Option 6 — a group of brand-new pan-KRAS drugs still in first-in-human testing

This option carries caveats. Here is the tradeoff to weigh. These are four very early drugs (BBO-11818, BLU-924, PF-07934040, and LY3962673) that block KRAS broadly and would cover your G12D mutation. They are on the page because they aim squarely at your main target and some have recruiting slots.

The honest problem is that there is almost nothing to go on yet. These are first-in-human studies, the earliest stage of testing, where the goal is mostly to find a safe dose. None of them has a real response rate published. The only public hint is a single patient on one of these drugs whose tumor shrank, with no larger group behind that one case, and no safety tables yet for any of the four. Ranking them any higher would be ranking novelty, not evidence. Think of these as worth keeping an eye on for later, with the first meaningful readout expected in the second half of this year, rather than as something to act on now.

Option 7 — ceralasertib (a pill aimed at the ARID1A loss, your second target)

This option carries caveats. Here is the tradeoff to weigh. This one aims at your other target, the ARID1A loss, through a completely different mechanism than the KRAS drugs. We keep it on the page because it is a genuinely independent second shot if the KRAS route does not work out. It is also the thinnest evidence here, with several real problems.

In the study that exists, across a mix of ARID1A-deficient cancers, the tumor shrank in about 14% of people, and the responses clustered in gynecologic cancers. No pancreatic cancer responded in that data. So applying it to your pancreatic cancer is an educated guess from the biology, not something shown to work in this disease.

Getting in is not even certain. The trial requires a specific tissue stain (BAF250a) to confirm the ARID1A protein is genuinely gone, and only about two-thirds of tumors with this gene change show that loss, so your genetic result alone does not guarantee eligibility. The main trial for it is currently not enrolling. The drug's class is also known for lowering blood counts, especially anemia, which is harder to tolerate at 79. This is here as a documented backup, not something to act on now.

A note on immune therapies and KRAS vaccines

You may have read about immune treatments and KRAS vaccines for this kind of cancer, so it is worth saying plainly why they are not on the list. The published KRAS-targeted cell therapy that made headlines only works in people with a particular immune-system type (a specific HLA marker), and your immune type is different, so it cannot recognize your tumor the way it needs to. The vaccine approaches that do not depend on immune type were studied in people whose cancer was caught early and largely removed by surgery, not in cancer that has spread the way yours has, so those results do not carry over. And the immune-checkpoint drugs are unlikely to help here, for the DNA-repair reasons above. None of this is a target your KRAS options are missing; those particular immune routes just do not fit your case.

Questions to ask your oncologist

- For Option 1 (daraxonrasib), what is the plan for catching and managing the skin rash and mouth sores early, and at what point would we cut the dose?
- The strongest evidence for Option 1 comes from people who were further along in treatment than I am. How much should we lean on it when I would be starting first-line?
- Given that I am 79, should we lean toward the gentler pill (Option 3) even though it has less evidence behind it? How do you weigh that for me specifically?
- Option 2 pairs its pill with an IV antibody, and Option 4 pairs its pill with chemotherapy. What extra side effects does each pairing add, and how would we watch for them?
- For Option 4, the combination caused a serious drop in white blood cells in about 40% of people. How would we monitor my blood counts, and what is the threshold for backing off?
- Option 5 had a very high rate of serious side effects and is not yet enrolling. Is there any scenario where it would be worth waiting for, or is it off the table for me?
- Can we get a tissue biopsy reviewed to confirm the KRAS finding, since most of these trials need that before I can enroll?
- Is it worth running the MMR protein stain on my tumor to settle the immune-therapy question one way or the other?
- For the ARID1A option (Option 7), can we order the confirming stain (BAF250a) now so we know whether it is even a real fallback, and are any trials for it actually open?

Sources

The recommendations on this page draw on the following published studies and trial records.

Published studies (PubMed ID): 27958275, 35648703, 36216931, 37625401, 38593348, 40790272, 41667470, 41686845, 42223072.

Conference abstracts and papers cited by DOI: 10.1016/j.annonc.2025.08.1484, 10.1016/j.annonc.2025.09.099, 10.1038/s41591-026-04538-9, 10.1200/JCO.2025.43.4_suppl.724, 10.1200/JCO.2026.44.2_suppl.654.

Clinical trials (ClinicalTrials.gov ID): NCT03682289, NCT06040541, NCT06179160, NCT06445062, NCT06447662, NCT06586515, NCT06625320, NCT06917079, NCT07232875, NCT07262567, NCT07491445, NCT07522073, NCT07621718, NCT07629960, NCT07644559.